

# Conversion with Periodic Repetition

Giannacco

Convert  $0.7_{10}$  to Binary

$$0.7 \times 2 = 1.4$$

$$\text{Remainder} = 1 = a_1$$

$$0.4 \times 2 = 0.8$$

$$\text{Remainder} = 0 = a_1$$

$$0.8 \times 2 = 1.6$$

$$\text{Remainder} = 1 = a_1$$

$$0.6 \times 2 = 1.2$$

$$\text{Remainder} = 1 = a_1$$

$$0.2 \times 2 = 0.4$$

$$\text{Remainder} = 0 = a_1$$

$$0.4 \times 2 = 0.8$$

$$\text{Remainder} = 0 = a_1$$

$$0.7_{10} = 0.10110011001100\dots_2$$